NONCOMPOSITE VALIDATION

$girder_span := 30\ ft$
$infill_span := 30\ ft$
$number_infills := 2$
$deck_depth := 2\ in$
$topping_depth := 2\ in$ (Concrete Topping Depth)
$SDL := 20\ psf$ (Superimposed Dead Load)
$LL := 40\ psf$ (Live Load)
occupancy=Residential

Validate Selected Deck

Selected Deck: 2" 22 Ga w/ 2" LWC
infill span is: 10' (30' / 3 equal spaces)
total slab depth: 4"
$load_on_floor := LL + SDL = 60\ psf$ which is < 139psf max for given design
10' span is also < 10'-2" max span for 3 clear spans which is assumed because bays are repeated
The 2" 22 Ga w/ 2" LWC deck selected in code is the lightest possible and passes

Dead Load

$avg_depth := 3\ in$ (Average Depth of Concrete)
$w_c := 110\ pcf$ (Density of Concrete)
$slab_SW := \frac{avg_depth}{12\ in}\ ft \cdot w_c = 27.5\ psf$ (Slab Self Weight)

$deck_SW := 1.6\ psf$ (Deck Self Weight)
$deck_slab_SW := slab_SW + deck_SW = 29.1\ psf$
$DL := slab_SW + deck_SW + SDL = 49.1\ psf$ (Dead Load)

Factored Loads

$LRFD_LC1 := 1.4 \cdot DL = 68.74\ psf$
$LRFD_LC2 := 1.2 \cdot DL + 1.6 \cdot LL = 122.92\ psf$
$ASD_LC1 := DL + LL = 89.1\ psf$

Infill Noncomposite Values

Selected Infill: W18x35
$E_s := 29000\ ksi$
$I_{x_i} := 510\ in^4$

$\phi V_{nx} := 159\ kip$
$\phi M_{px} := 249\ kip \cdot ft$
(Both values are taken from Table 3-2 as it is assumed the infills are fully braced against the deck)

$infill_spacing := \frac{girder_span}{number_infills + 1} = 10\ ft$

Infill Calculated Values

$infill_SW := 35\ plf$ (Infill Self Weight)

$w_{DL_i} := DL \cdot infill_spacing + infill_SW = 0.526\ klf$ (Infill DL Uniform Load)

$w_{LL_i} := LL \cdot infill_spacing = 0.4\ klf$ (Infill LL Uniform Load)

$w_{FL_i} := 1.2 \cdot w_{DL_i} + 1.6 \cdot w_{LL_i} = 1.271\ klf$ (Infill Factored Uniform Load)

$$V_{DL_i} := \frac{w_{DL_i} \cdot infill_span}{2} = 7.89\ kip$$

$$V_{LL_i} := \frac{w_{LL_i} \cdot infill_span}{2} = 6\ kip$$

$$V_{FL_i} := \frac{w_{FL_i} \cdot infill_span}{2} = 19.068\ kip \qquad V_{FL_i} < 159\ kip$$

$$M_{FL_i} := \frac{w_{FL_i} \cdot infill_span^2}{8} = 143.01\ kip \cdot ft \qquad M_{FL_i} < 249\ kip \cdot ft$$

$$\Delta_{LL} := \frac{5 \cdot w_{LL_i} \cdot infill_span^4}{384 \cdot E_s \cdot I_{x_i}} = 0.4929\ in \qquad \Delta_{LL} < 1\ in \qquad l/360$$

$$\Delta_{DL} := \frac{5 \cdot w_{DL_i} \cdot infill_span^4}{384 \cdot E_s \cdot I_{x_i}} = 0.6482\ in$$

$\Delta_{TL} := \Delta_{LL} + \Delta_{DL} = 1.1411\ in$ $\qquad \Delta_{TL} < 1.5\ in$ $\qquad l/240$

Girder Noncomposite Values

Selected Infill: W21X55

$I_{x_g} := 1140\ in^4$

$A_g := 16.2\ in^2$

$b_{f_g} := 8.22\ in$

$d_g := 20.8\ in$

$L_b := 10\ ft$

$C_b := 1.0$ Cb=1.0 for bays with 2 equally spaced infills

$L_p := 6.11\ ft$ Because Lb is between Lp and Lr the member is

$L_r := 13.9\ ft$ noncompact and LTB is checked

$\phi M_{px} := 473\ kip \cdot ft$

$\phi BF := 16.3\ kip$

$\phi M_{n_g} := C_b \cdot (\phi M_{px} - \phi BF \cdot (L_b - L_p)) = 409.593\ kip \cdot ft$

$\phi V_{nx} := 234\ kip$ (from Table 3-2)

Girder Calculated Values

$girder_SW := 55\ plf$ (Girder Self Weight)

$$V_{DL_g} := \frac{\left(girder_SW \cdot girder_span + 2 \cdot V_{DL_i} \cdot number_infills\right)}{2} = 16.605\ kip$$

$$V_{LL_g} := \frac{2 \cdot V_{LL_i} \cdot number_infills}{2} = 12\ kip$$

$V_{FL_g} := 1.2 \cdot V_{DL_g} + 1.6 \cdot V_{LL_g} = 39.126\ kip$ $V_{FL_g} < 234\ k$

$$M_{SW_g} := \frac{girder_SW \cdot girder_span^2}{8} = 6.188\ kip \cdot ft$$

$M_{DL_g} := M_{SW_g} + 2\ V_{DL_i} \cdot infill_spacing = 163.988\ kip \cdot ft$

$M_{LL_g} := 2\ V_{LL_i} \cdot infill_spacing = 120\ kip \cdot ft$

$M_{FL_g} := 1.2 \cdot M_{DL_g} + 1.6 \cdot M_{LL_g} = 388.785\ kip \cdot ft$ $M_{FL_g} < 409.593\ kip \cdot ft$

$$\Delta_{SW_g} := \frac{5 \cdot girder_SW \cdot girder_span^4}{384 \cdot E_s \cdot I_{x_g}} = 0.0303\ in$$

$$\Delta_{DL_g} := \Delta_{SW_g} + 2 \cdot \frac{23 \cdot V_{DL_i} \cdot girder_span^3}{648 \cdot E_s \cdot I_{x_g}} = 0.8208\ in$$

$$\Delta_{LL_g} := 2 \cdot \frac{23 \cdot V_{LL_i} \cdot girder_span^3}{648 \cdot E_s \cdot I_{x_g}} = 0.6011\ in$$

$\Delta_{comp_LL_g} < \frac{girder_span}{360}$ or 1"

$\Delta_{TL_g} := \Delta_{LL_g} + \Delta_{DL_g} = 1.4218\ in$

$\Delta_{comp_TL_g} < \frac{girder_span}{240}$ or 1.5"

Vibration (Peak Acceleration) Calculation

$A_i := 10.3\ in^2$

$d_i := 17.7\ in$

$f'_c := 4\ ksi$

$E_c := \left(\frac{w_c}{pcf}\right)^{1.5} \cdot \sqrt[2]{\frac{f'_c}{ksi}} = 2.307 \cdot 10^3$ $\quad E_c := 2307.38\ ksi$

$LL_{vib} := 6\ psf$

$DL_{vib} := 4\ psf$

$\beta := 0.02$

$n := \frac{E_s}{1.35 \cdot E_c} = 9.31$

$conc_width_{eff_i} := min\left(0.4 \cdot infill_span, infill_spacing\right) = 120\ in$

$conc_width_{tran_i} := \frac{conc_width_{eff_i}}{n} = 12.89\ in$

$conc_depth_{eff} := topping_depth = 2\ in$

$conc_area_{tran_i} := conc_depth_{eff} \cdot conc_width_{tran_i} = 25.779\ in^2$

$y_{bar_i} := \frac{conc_area_{tran_i} \cdot \left(\frac{d_i}{2} + deck_depth + \frac{conc_depth_{eff}}{2}\right)}{conc_area_{tran_i} + A_i} = 8.467\ in$

$I_i := \frac{conc_width_{tran_i} \cdot conc_depth_{eff}^3}{12} + conc_area_{tran_i} \cdot \left(\frac{d_i}{2} + deck_depth + \frac{conc_depth_{eff}}{2} - y_{bar_i}\right)^2 + I_{x_i} + A_i \cdot y_{bar_i}^2 = \left(1.552 \cdot 10^3\right)\ in^4$

$w_i := infill_spacing \cdot \left(LL_{vib} + deck_slab_SW + DL_{vib}\right) + infill_SW = 426\ plf$

$\Delta_i := \frac{5 \cdot w_i \cdot infill_span^4}{384 \cdot E_s \cdot I_i} = 0.172\ in$

$g := 386\ \frac{in}{s^2}$

$f_i := 0.18 \cdot \sqrt[2]{\frac{g}{\Delta_i}} = 8.515\ Hz$

$d_e := topping_depth + \frac{deck_depth}{2} = 3\ in$

$$D_s := \frac{\left(\frac{12\ in}{1\ ft}\right) d_e^{\,3}}{12\ n} = 2.9\ \frac{in^4}{ft}$$

$$D_i := \frac{I_i}{infill_spacing} = 155.203\ \frac{in^4}{ft}$$

$$C_j := 2.0$$

$$B_i := C_j \cdot \left(\frac{D_s}{D_i}\right)^{0.25} \cdot infill_span = 22.184\ ft$$

floor width and length are assumed to be large enough to not cause any problems

$$W_i := 1.5 \cdot \left(\frac{w_i}{infill_spacing}\right) \cdot B_i \cdot infill_span = 42525.8\ lbf$$

$$conc_width_{eff_g} := 2 \cdot min\left(0.2 \cdot girder_span, 0.5 \cdot infill_span\right) = 144\ in$$

$$conc_width_{tran_g} := \frac{conc_width_{eff_g}}{n} = 15.467\ in$$

$$deck_width_{eff_g} := \frac{conc_width_{eff_g}}{2\ n} = 7.734\ in$$

$$conc_area_{tran_g} := conc_depth_{eff} \cdot conc_width_{tran_g} = 30.935\ in^2$$

$$deck_area_{tran_g} := deck_depth \cdot deck_width_{eff_g} = 15.467\ in^2$$

$$y_{bar_g} := \frac{conc_area_{tran_g} \cdot \left(\frac{d_g}{2} + deck_depth + \frac{conc_depth_{eff}}{2}\right) + deck_area_{tran_g} \cdot \left(\frac{d_g}{2} + \frac{deck_depth}{2}\right)}{conc_area_{tran_g} + deck_area_{tran_g} + A_g} = 9.44\ in$$

$$I_g := \frac{conc_width_{tran_g} \cdot conc_depth_{eff}^{\,3}}{12} + conc_area_{tran_g} \cdot \left(\frac{d_g}{2} + deck_depth + \frac{conc_depth_{eff}}{2} - y_{bar_g}\right)^2 + \frac{deck_width_{eff_g} \cdot deck_depth^3}{12} + deck_area_{tran_g} \cdot \left(\frac{d_g}{2} + \frac{deck_depth}{2} - y_{bar_g}\right)^2 + I_{x_g} + A_g \cdot y_{bar_g}^{\,2} = 3143.6\ in^4$$

$$w_g := infill_span \cdot \left(\frac{w_i}{infill_spacing}\right) + girder_SW = 1333\ plf$$

$$\Delta_g := \frac{5 \cdot w_g \cdot girder_span^4}{384 \cdot E_s \cdot I_g} = 0.266\ in$$

$$f_g := 0.18 \cdot \sqrt{\frac{g}{\Delta_g}} = 6.851\ Hz$$

$$D_g := \frac{I_g}{infill_span} = 104.788\ \frac{in^4}{ft}$$

$$C_g := 1.8$$

$$B_g := C_g \cdot \left(\frac{D_i}{D_g}\right)^{0.25} \cdot girder_span = 59.572\ ft$$

$$W_g := \left(\frac{w_g}{infill_span}\right) \cdot B_g \cdot infill_span = 79409.3\ lbf$$

$$f_n := 0.18 \cdot \sqrt{\frac{g}{\Delta_i + \Delta_g}} = 5.338\ Hz$$

$$\Delta_g' := \Delta_g = 0.266\ in$$ Girder Span is > B_i

$$W := \frac{\Delta_i}{\Delta_i + \Delta_g'} \cdot W_i + \frac{\Delta_g}{\Delta_i + \Delta_g'} \cdot W_g = 64916\ lbf$$

$$P_o := 65\ lbf$$

$$a_{p_over_g} := \left(\frac{P_o \cdot e^{-0.35 \cdot \frac{f_n}{Hz}}}{\beta \cdot W}\right) \cdot 100 = 0.77304\ \%g$$

Embodied Carbon Calculation

$bay_area := girder_span \cdot infill_span = 900\ ft^2$

$W_{infill} := infill_SW \cdot infill_span \cdot (number_infills + 2) = 4200\ lbf$

$W_{girder} := girder_SW \cdot girder_span \cdot 2 = 3300\ lbf$

$W_{deck} := deck_SW \cdot bay_area = 1440\ lbf$

$W_{per_stud} := 0.437\ lbf$

$W_{concrete} := slab_SW \cdot bay_area = 24750\ lbf$

$EC_{steel_members} := 1.22 \cdot \left(\frac{W_{infill} + W_{girder}}{lbf} \right) = 9150$ lbCO2e/lb

$EC_{deck} := 2.32 \cdot \left(\frac{W_{deck}}{lbf} \right) = 3340.8$ lbCO2e/lb

$EC_{concrete} := 0.307 \cdot \left(\frac{W_{concrete}}{lbf} \right) = 7598.3$ lbCO2e/lb

$Scaled_EC := \frac{EC_{steel_members} + EC_{deck} + EC_{concrete}}{\frac{bay_area}{ft^2}} = 22.321$ lbCO2e/(lb*ft^2)

Spot Checks: Checking Slender Moment Capacity (Lr < Lb)

* For any of the 'Spot Checks', values highlighted in green match those output in code

To achieve slender conditions, the following design was used:
Girder Span: 30'
Infill Span: 15'
Equally Spaced Infills: 1
Concrete Type: LWC
Deck Depth: 3"
Occupancy: Residential

* Only the flexural capacity of the W18x40 Girder selected in the code is checked to ensure Slender Moment Capacity is calculated properly

$L_b := \frac{30\ ft}{2} = 15\ ft$ (2 equal spaces)

$L_r := 13.1\ ft$

$L_b > L_r$ so the girder is slender

$C_b := 1.67$ (2 equal spaces Table 3-1)
$r_{ts} := 1.56\ in$
$J := 0.81\ in^4$
$S_x := 68.4\ in^3$
$h_o := 17.4\ in$

$$F_{cr} := \frac{C_b \cdot \pi^2 \cdot E_s}{\left(\frac{L_b}{r_{ts}}\right)^2} \cdot \sqrt{1 + 0.078 \cdot \frac{J}{S_x \cdot h_o}\left(\frac{L_b}{r_{ts}}\right)^2} = 46.903\ ksi$$

$\phi M_n := 0.9 \cdot S_x \cdot F_{cr} = 240.614\ kip \cdot ft$ (Matches value calculated in code)

Spot Checks: Making sure compact members aren't checked for LTB (Lp > Lb)

To achieve compact conditions, the following design was used:
Girder Span: 10'
Infill Span: 15'
Equally Spaced Infills: 3
Concrete Type: LWC
Deck Depth: 3"
Occupancy: Residential

* It is understood this is an unrealistic design, it is used to ensure the boundaries of my design space are being calculated correctly
* Only the flexural capacity of the W10x12 Girder selected in the code is checked to ensure Compact Moment Capacity is calculated properly

$L_b := \frac{10\ ft}{4} = 2.5\ ft$ (4 equal spaces)

$L_p := 2.87\ ft$

$L_b < L_p$ so the girder is compact, and PhiMpx from Table 3-2 is used

$\phi M_n := 47\ kip \cdot ft$ (Matches value used in code)

Spot Checks: Shear, Moment, and Deflection Values for One Infill

The following design was used:
Girder Span: 30'
Infill Span: 30'
Equally Spaced Infills: 1
Concrete Type: LWC
Deck Depth: 3"
Occupancy: Residential

$L_i := 30\ ft$
$L_g := 30\ ft$

$LL := 40\ psf$
$SDL := 20\ psf$
$infill_spacing := \frac{30\ ft}{2} = 15\ ft$

3" Deck 19Ga w/ 2" LWC
$Deck_Slab_SW := 34.5\ psf$
$DL := SDL + Deck_Slab_SW = 54.5\ psf$

Infill: W18x40, Girder: W21x55
$Infill_SW := 40\ plf$
$Girder_SW := 55\ plf$

Infill Values
$w_{DL} := DL \cdot infill_spacing + Infill_SW = 0.8575\ klf$
$w_{LL} := LL \cdot infill_spacing = 0.6\ klf$
$w_{FL} := 1.2 \cdot w_{DL} + 1.6 \cdot w_{LL} = 1.989\ klf$

$V_{LL} := \frac{w_{LL} \cdot L_i}{2} = 9\ kip$

$V_{DL} := \frac{w_{DL} \cdot L_i}{2} = 12.863\ kip$

$V_{FL} := \frac{w_{FL} \cdot L_i}{2} = 29.835\ kip$

$M_{FL} := \frac{w_{FL} \cdot {L_i}^2}{8} = 223.763\ kip \cdot ft$

$I_x := 612\ in^4$

$\Delta_{LL} := \frac{5 \cdot w_{LL} \cdot L_i^4}{384 \cdot E_s \cdot I_x} = 0.6161\ in$

$\Delta_{DL} := \frac{5 \cdot w_{DL} \cdot L_i^4}{384 \cdot E_s \cdot I_x} = 0.8805\ in$

$\Delta_{TL} := \Delta_{LL} + \Delta_{DL} = 1.497\ in$

Girder Values

$V_{DL_g} := \frac{Girder_SW \cdot L_g}{2} + 2 \cdot \frac{V_{DL}}{2} = 13.688\ kip$

(Superimpose linear SW with half shear of one infill from two adjacent bays)

$V_{LL_g} := 2 \cdot \frac{V_{LL}}{2} = 9\ kip$

$V_{FL_g} := 1.2 \cdot V_{DL_g} + 1.6 \cdot V_{LL_g} = 30.825\ kip$

$M_{DL_g} := \frac{Girder_SW \cdot L_g^2}{8} + 2 \cdot \frac{V_{DL} \cdot L_g}{4} = 199.125\ kip \cdot ft$

$M_{LL_g} := 2 \cdot \frac{V_{LL} \cdot L_g}{4} = 135\ kip \cdot ft$

$M_{FL_g} := 1.2 \cdot M_{DL_g} + 1.6 \cdot M_{LL_g} = 454.95\ ft \cdot kip$

$I_{x_g} := 1140\ in^4$

$\Delta_{LL_g} := 2\ \frac{V_{LL} \cdot L_g^3}{48 \cdot E_s \cdot I_{x_g}} = 0.5292\ in$

$\Delta_{DL_g} := \frac{5 \cdot Girder_SW \cdot L_g^4}{384 \cdot E_s \cdot I_{x_g}} + 2 \cdot \frac{V_{DL} \cdot L_g^3}{48 \cdot E_s \cdot I_{x_g}} = 0.7867\ in$

$\Delta_{TL_g} := \Delta_{LL_g} + \Delta_{DL_g} = 1.316\ in$

Spot Checks: Shear, Moment, and Deflection Values for Three Infills

The following design was used:
Girder Span: 30'
Infill Span: 30'
Equally Spaced Infills: 3
Concrete Type: LWC
Deck Depth: 3"
Occupancy: Residential

$L_i := 30\ ft$
$L_g := 30\ ft$

$LL := 40\ psf$
$SDL := 20\ psf$
$infill_spacing := \frac{30\ ft}{4} = 7.5\ ft$

3" Deck 22Ga w/ 2" LWC
$Deck_Slab_SW := 33.8\ psf$
$DL := SDL + Deck_Slab_SW = 53.8\ psf$

Infill: W16x31, Girder: W24x55
$Infill_SW := 31\ plf$
$Girder_SW := 55\ plf$

Infill Values
$w_{DL} := DL \cdot infill_spacing + Infill_SW = 0.4345\ klf$
$w_{LL} := LL \cdot infill_spacing = 0.3\ klf$
$w_{FL} := 1.2 \cdot w_{DL} + 1.6 \cdot w_{LL} = 1.001\ klf$

$V_{LL} := \frac{w_{LL} \cdot L_i}{2} = 4.5\ kip$

$V_{DL} := \frac{w_{DL} \cdot L_i}{2} = 6.518\ kip$

$V_{FL} := \frac{w_{FL} \cdot L_i}{2} = 15.021\ kip$

$M_{FL} := \frac{w_{FL} \cdot L_i^{\,2}}{8} = 112.658\ kip \cdot ft$

$I_x := 375\ in^4$

$\Delta_{LL} := \frac{5 \cdot w_{LL} \cdot L_i^4}{384 \cdot E_s \cdot I_x} = 0.5028\ in$

$\Delta_{DL} := \frac{5 \cdot w_{DL} \cdot L_i^4}{384 \cdot E_s \cdot I_x} = 0.7282\ in$

$\Delta_{TL} := \Delta_{LL} + \Delta_{DL} = 1.231\ in$

Girder Values

$V_{DL_g} := \frac{Girder_SW \cdot L_g}{2} + 2 \cdot \frac{V_{DL} \cdot 3}{2} = 20.378\ kip$ (Superimpose linear SW with half shear of three infills from two adjacent bays)

$V_{LL_g} := 2 \cdot \frac{V_{LL} \cdot 3}{2} = 13.5\ kip$

$V_{FL_g} := 1.2 \cdot V_{DL_g} + 1.6 \cdot V_{LL_g} = 46.053\ kip$

$M_{DL_g} := \frac{Girder_SW \cdot L_g^2}{8} + 2 \cdot \frac{V_{DL} \cdot L_g}{2} = 201.713\ kip \cdot ft$

$M_{LL_g} := 2 \cdot \frac{V_{LL} \cdot L_g}{2} = 135\ kip \cdot ft$

$M_{FL_g} := 1.2 \cdot M_{DL_g} + 1.6 \cdot M_{LL_g} = 458.055\ ft \cdot kip$

$I_{x_g} := 1350\ in^4$

$\Delta_{LL_g} := 2\ \frac{19\ V_{LL} \cdot L_g^3}{384 \cdot E_s \cdot I_{x_g}} = 0.5307\ in$

$\Delta_{DL_g} := \frac{5 \cdot Girder_SW \cdot L_g^4}{384 \cdot E_s \cdot I_{x_g}} + 2 \cdot \frac{19\ V_{DL} \cdot L_g^3}{384 \cdot E_s \cdot I_{x_g}} = 0.7942\ in$

$\Delta_{TL_g} := \Delta_{LL_g} + \Delta_{DL_g} = 1.325\ in$

Spot Checks: Shear, Moment, and Deflection Values for Four Infills

The following design was used:
Girder Span: 30'
Infill Span: 30'
Equally Spaced Infills: 4
Concrete Type: LWC
Deck Depth: 3"
Occupancy: Residential

$L_i := 30\ ft$
$L_g := 30\ ft$

$LL := 40\ psf$
$SDL := 20\ psf$
$infill_spacing := \frac{30\ ft}{5} = 6\ ft$

3" Deck 22Ga w/ 2" LWC
$Deck_Slab_SW := 33.8\ psf$
$DL := SDL + Deck_Slab_SW = 53.8\ psf$

Infill: W16x26, Girder: W24x55
$Infill_SW := 26\ plf$
$Girder_SW := 55\ plf$

Infill Values
$w_{DL} := DL \cdot infill_spacing + Infill_SW = 0.3488\ klf$
$w_{LL} := LL \cdot infill_spacing = 0.24\ klf$
$w_{FL} := 1.2 \cdot w_{DL} + 1.6 \cdot w_{LL} = 0.803\ klf$

$V_{LL} := \frac{w_{LL} \cdot L_i}{2} = 3.6\ kip$

$V_{DL} := \frac{w_{DL} \cdot L_i}{2} = 5.232\ kip$

$V_{FL} := \frac{w_{FL} \cdot L_i}{2} = 12.038\ kip$

$M_{FL} := \frac{w_{FL} \cdot L_i^{\,2}}{8} = 90.288\ kip \cdot ft$

$I_x := 301\ in^4$

$\Delta_{LL} := \frac{5 \cdot w_{LL} \cdot L_i^4}{384 \cdot E_s \cdot I_x} = 0.5011\ in$

$\Delta_{DL} := \frac{5 \cdot w_{DL} \cdot L_i^4}{384 \cdot E_s \cdot I_x} = 0.7282\ in$

$\Delta_{TL} := \Delta_{LL} + \Delta_{DL} = 1.229\ in$

Girder Values

$w_{DL_g} := DL \cdot L_i = 1.614\ klf$

$w_{LL_g} := LL \cdot L_i = 1.2\ klf$

(Infill Span is spacing between girders)

$w_{SW_i} := \frac{4 \cdot Infill_SW \cdot L_i}{L_g}$

(Self weight of all infills resting on girder)

$V_{DL_g} := \frac{Girder_SW \cdot L_g}{2} + 2 \cdot \frac{V_{DL} \cdot 4}{2} = 21.753\ kip$

$V_{LL_g} := 2 \cdot \frac{V_{LL} \cdot 4}{2} = 14.4\ kip$

$V_{FL_g} := 1.2 \cdot V_{DL_g} + 1.6 \cdot V_{LL_g} = 49.144\ kip$

$M_{DL_g} := \frac{(Girder_SW + w_{SW_i} + w_{DL_g}) \cdot L_g^2}{8} = 199.463\ kip \cdot ft$

$M_{LL_g} := \frac{w_{LL_g} \cdot L_g^2}{8} = 135\ kip \cdot ft$

(Moment and Deflection values conservatively assumed to be uniform for simplicity for more than 3 evenly spaced infills)

$M_{FL_g} := 1.2 \cdot M_{DL_g} + 1.6 \cdot M_{LL_g} = 455.355\ ft \cdot kip$

$I_{x_g} := 1350\ in^4$

$\Delta_{LL_g} := \frac{5 \cdot (w_{LL_g}) \cdot L_g^4}{384 \cdot E_s \cdot I_{x_g}} = 0.5586\ in$

$\Delta_{DL_g} := \frac{5 \cdot (Girder_SW + w_{SW_i} + w_{DL_g}) \cdot L_g^4}{384 \cdot E_s \cdot I_{x_g}} = 0.8254\ in$

$\Delta_{TL_g} := \Delta_{LL_g} + \Delta_{DL_g} = 1.384\ in$

COMPOSITE VALIDATION

$girder_span := 30\ ft$
$infill_span := 30\ ft$
$number_infills := 2$
$deck_depth := 2\ in$
$topping_depth := 2\ in$ (Concrete Topping Depth)
$PCC := 50\%$ (Percent Composite Connection)
$SDL := 20\ psf$ (Superimposed Dead Load)
$LL := 40\ psf$ (Live Load)
occupancy=Residential

Validate Selected Deck

Selected Deck: 2" 22 Ga w/ 2" LWC
infill span is: 10' (30' / 3 equal spaces)
total slab depth: 4"
$load_on_floor := LL + SDL = 60\ psf$ which is < 139psf max for given design
10' span is also < 10'-2" max span for 3 clear spans which is assumed because bays are repeated
The 2" 22 Ga w/ 2" LWC deck selected in code is the lightest possible and passes

Dead Load

$avg_depth := 3\ in$ (Average Depth of Concrete)
$w_c := 110\ pcf$ (Density of Concrete)
$slab_SW := \frac{avg_depth}{12\ in}\ ft \cdot w_c = 27.5\ psf$ (Slab Self Weight)

$deck_SW := 1.6\ psf$ (Deck Self Weight)
$deck_slab_SW := slab_SW + deck_SW = 29.1\ psf$
$DL := slab_SW + deck_SW + SDL = 49.1\ psf$ (Dead Load)

Factored Loads

$LRFD_LC1 := 1.4 \cdot DL = 68.74\ psf$
$LRFD_LC2 := 1.2 \cdot DL + 1.6 \cdot LL = 122.92\ psf$
$ASD_LC1 := DL + LL = 89.1\ psf$

Infill Composite Values

Selected Infill: W12x19 [17]
$E_s := 29000\ ksi$
$I_{x_i} := 130\ in^4$
$f'_c := 4\ ksi$
$A_i := 5.57\ in^2$
$F_y := 50\ ksi$
$F_u := 65\ ksi$
$b_{f_i} := 4.01\ in$
$d_i := 12.2\ in$

$R_g := 1$

$R_p := 0.6$

$A_{stud} := \frac{\pi \cdot (0.75\ in)^2}{4} = 0.442\ in^2$

$E_c := \left(\frac{w_c}{pcf}\right)^{1.5} \cdot \sqrt[2]{\frac{f'_c}{ksi}} = 2.307 \cdot 10^3$ $\quad E_c := 2307.38\ ksi$

$infill_spacing := \frac{girder_span}{number_infills + 1} = 10\ ft$

$b_{eff_i} := min\left(\frac{infill_span \cdot 2}{8}, \frac{infill_spacing \cdot 2}{2}\right) = 90\ in$

$V_{c_i}' := 0.85 \cdot f'_c \cdot b_{eff_i} \cdot topping_depth = 612\ kip$

$V_{s_i}' := A_i \cdot F_y = 278.5\ kip$

$V_i' := min\left(V_{c_i}', V_{s_i}'\right) = 278.5\ kip$

$Q_{n_i} := min\left(0.5 \cdot A_{stud} \cdot \sqrt[2]{(f'_c \cdot E_c)}, R_g \cdot R_p \cdot A_{stud} \cdot F_u\right) = 17.23\ kip$

$\Sigma Q_{n_i} := PCC \cdot V_{s_i}' = 139.25\ kip$ 17 studs are specified, therefore 8.5 studs * Qn = 146kip > 139kip

$a_i := \frac{\Sigma Q_{n_i}}{0.85 \cdot f'_c \cdot b_{eff_i}} = 0.455\ in$

$C_{c_i} := 0.85 \cdot f'_c \cdot b_{eff_i} \cdot a_i = 139.25\ kip$ $\quad t_{f_i} := 0.350\ in$

$Y1_i := \frac{A_i \cdot F_y - C_{c_i}}{2 \cdot b_{f_i} \cdot F_y} = 0.347\ in$ Y1 < tf, PNA is in the Steel Beam Flange

$Y2_i := deck_depth + topping_depth - \frac{a_i}{2} = 3.772\ in$

$T_i := A_i \cdot F_y = 278.5\ kip$

$\phi M_{n_i} := 0.9 \cdot \left(C_{c_i} \cdot \left(Y2_i + \frac{Y1_i}{2}\right) + T_i \cdot \left(\frac{d_i}{2} - \frac{Y1_i}{2}\right)\right) = 164.999\ kip \cdot ft$

$d_{1_i} := topping_depth + deck_depth - \frac{a_i}{2} = 3.772\ in$

$d_{3_i} := \frac{d_i}{2} = 6.1\ in$

$Y_{ENA_i} := \frac{\left(A_i \cdot d_{3_i} + \frac{\Sigma Q_{n_i}}{F_y} \cdot (2 \cdot d_{3_i} + d_{1_i})\right)}{\left(A_i + \frac{\Sigma Q_{n_i}}{F_y}\right)} = 9.391\ in$

$I_{LB_i} := I_{x_i} + A_i \cdot (Y_{ENA_i} - d_{3_i})^2 + \frac{\Sigma Q_{n_i}}{F_y} \cdot (2 \cdot d_{3_i} + d_{1_i} - Y_{ENA_i})^2 = 310.961\ in^4$

Infill Calculated Values

$infill_SW := 19\ plf$ (Infill Self Weight)

$w_{DL_i} := DL \cdot infill_spacing + infill_SW = 0.51\ klf$ (Infill DL Uniform Load)

$w_{LL_i} := LL \cdot infill_spacing$ (Infill LL Uniform Load)

$w_{FL_i} := 1.2 \cdot w_{DL_i} + 1.6 \cdot w_{LL_i} = 1.252\ klf$ (Infill Factored Uniform Load)

(Precomposite Factored Uniform Load)

$w_{precomp_i} := 1.2 \cdot (deck_slab_SW \cdot infill_spacing + 19\ plf) + 1.6\ (20\ psf \cdot infill_spacing) = 0.692\ klf$

$w_{con_i} := deck_slab_SW \cdot infill_spacing + 19\ plf = ?\ klf$

$w_{comp_LL_i} := LL \cdot infill_spacing \cdot 0.5 = 0.2\ klf$

$$V_{DL_i} := \frac{w_{DL_i} \cdot infill_span}{2} = 7.65\ kip$$

$$V_{LL_i} := \frac{w_{LL_i} \cdot infill_span}{2} = 6\ kip$$

$$V_{FL_i} := \frac{w_{FL_i} \cdot infill_span}{2} = 18.78\ kip \qquad V_{FL_i} < 86\ kip$$

$$V_{con_i} := \frac{w_{con_i} \cdot infill_span}{2} = 4.65\ kip$$

$$V_{precomp_i} := \frac{w_{precomp_i} \cdot infill_span}{2} = 10.38\ kip$$

$$V_{comp_LL_i} := \frac{w_{comp_LL_i} \cdot infill_span}{2} = 3\ kip$$

$$M_{FL_i} := \frac{w_{FL_i} \cdot infill_span^2}{8} = 140.85\ kip \cdot ft \qquad M_{FL_i} < 165\ kip \cdot ft$$

$$M_{precomp_i} := \frac{w_{precomp_i} \cdot infill_span^2}{8} = 77.85\ kip \cdot ft \qquad M_{precomp_i} < 92.6\ kip \cdot ft$$

$$\Delta_{con_i} := \frac{5 \cdot w_{con_i} \cdot infill_span^4}{384 \cdot E_s \cdot I_{x_i}} = 1.4986\ in$$

$camber := 0.8 \cdot \Delta_{con_i} = 1.199\ in$

$camber_i := 1.25\ in$ gets rounded to nearest 1/4"

$$\Delta_{comp_LL_i} := \frac{5 \cdot w_{comp_LL_i} \cdot infill_span^4}{384 \cdot E_s \cdot I_{LB_i}} = 0.4042\ in \qquad \Delta_{comp_LL_g} < \frac{girder_span}{360} \text{ or } 1"$$

$$\Delta_{comp_DL_i} := \Delta_{con_i} - camber_i = 0.2486\ in \qquad \Delta_{comp_DL_g} < \frac{girder_span}{360} \text{ or } 1"$$

$$\Delta_{comp_TL_i} := \Delta_{comp_LL_i} + \Delta_{comp_DL_i} = 0.6528\ in \qquad \Delta_{comp_TL_g} < \frac{girder_span}{240} \text{ or } 1.5"$$

Girder Precomposite Values

Selected Infill: W18x40 [41]

$I_{x_g} := 612\ in^4$

$A_g := 11.8\ in^2$

$b_{f_g} := 6.02\ in$

$d_g := 17.9\ in$

$L_b := 10\ ft$

$C_b := 1.0$ Cb=1.0 for bays with 2 equally spaced infills

$L_p := 4.49\ ft$ Because Lb is between Lp and Lr the member is

$L_r := 13.1\ ft$ noncompact and LTB is checked

$\phi M_{px} := 294\ kip \cdot ft$

$\phi BF := 13.3\ kip$

$\phi M_{n_nc_g} := C_b \cdot \langle \phi M_{px} - \phi BF \cdot \langle L_b - L_p \rangle \rangle = 220.717\ kip \cdot ft$

$\phi V_{nx} := 169\ kip$ (from Table 3-2)

Girder Composite Values

$R_{g_g} := 0.85$ 2 studs per rib decreases Rg

$b_{eff_g} := min\left(\frac{girder_span \cdot 2}{8}, \frac{infill_span \cdot 2}{2}\right) = 90\ in$

$V_{c_g}' := 0.85 \cdot f'_c \cdot b_{eff_g} \cdot topping_depth = 612\ kip$
$V_{s_g}' := A_g \cdot F_y = 590\ kip$
$V_g' := min\left(V_{c_g}', V_{s_g}'\right) = 590\ kip$

$Q_{n_g} := min\left(0.5 \cdot A_{stud} \cdot \sqrt[2]{\left(f'_c \cdot E_c\right)}, R_{g_g} \cdot R_p \cdot A_{stud} \cdot F_u\right) = 14.645\ kip$

$\Sigma Q_{n_g} := PCC \cdot V_{s_g}' = 295\ kip$ 41 studs are specified, therefore 20.5 studs * Qn = 300kip > 295kip

$a_g := \frac{\Sigma Q_{n_g}}{0.85 \cdot f'_c \cdot b_{eff_g}} = 0.964\ in$

$C_{c_g} := 0.85 \cdot f'_c \cdot b_{eff_g} \cdot a_g = 295\ kip$ $t_{f_g} := 0.525\ in$

$Y1_g := \frac{A_g \cdot F_y - C_{c_g}}{2 \cdot b_{f_g} \cdot F_y} = 0.49\ in$ Y1 < tf, PNA is in the Steel Beam Flange

$Y2_g := deck_depth + topping_depth - \frac{a_g}{2} = 3.518\ in$
$T_g := A_g \cdot F_y = 590\ kip$

$\phi M_{n_g} := 0.9 \cdot \left(C_{c_g} \cdot \left(Y2_g + \frac{Y1_g}{2}\right) + T_g \cdot \left(\frac{d_g}{2} - \frac{Y1_g}{2}\right)\right) = 468.452\ kip \cdot ft$

$d_{1_g} := topping_depth + deck_depth - \frac{a_g}{2} = 3.518\ in$

$d_{3_g} := \frac{d_g}{2} = 8.95\ in$

$Y_{ENA_g} := \frac{\left(A_g \cdot d_{3_g} + \frac{\Sigma Q_{n_g}}{F_y} \cdot \left(2 \cdot d_{3_g} + d_{1_g}\right)\right)}{\left(A_g + \frac{\Sigma Q_{n_g}}{F_y}\right)} = 13.106\ in$

$I_{LB_g} := I_{x_g} + A_g \cdot \left(Y_{ENA_g} - d_{3_g}\right)^2 + \frac{\Sigma Q_{n_g}}{F_y} \cdot \left(2 \cdot d_{3_g} + d_{1_g} - Y_{ENA_g}\right)^2 = 1223.4\ in^4$

Girder Calculated Values

$girder_SW := 40\ plf$ (Girder Self Weight)

$V_{DL_g} := \frac{(girder_SW \cdot girder_span + 2 \cdot V_{DL_i} \cdot number_infills)}{2} = 15.9\ kip$

$V_{LL_g} := \frac{2 \cdot V_{LL_i} \cdot number_infills}{2} = 12\ kip$

$V_{FL_g} := 1.2 \cdot V_{DL_g} + 1.6 \cdot V_{LL_g} = 38.28\ kip$ $V_{FL_g} < 169\ k$

$V_{precomp_g} := \frac{(girder_SW \cdot girder_span + 2 \cdot V_{precomp_i} \cdot number_infills)}{2} = 21.36\ kip$

$V_{con_g} := \frac{(girder_SW \cdot girder_span + 2 \cdot V_{con_i} \cdot number_infills)}{2} = 9.9\ kip$

$M_{SW_g} := \frac{girder_SW \cdot girder_span^2}{8} = 4.5\ kip \cdot ft$

$M_{DL_g} := M_{SW_g} + 2\ V_{DL_i} \cdot infill_spacing = 157.5\ kip \cdot ft$

$M_{LL_g} := 2\ V_{LL_i} \cdot infill_spacing = 120\ kip \cdot ft$

$M_{FL_g} := 1.2 \cdot M_{DL_g} + 1.6 \cdot M_{LL_g} = 381\ kip \cdot ft$ $M_{FL_g} < 468.5\ kip \cdot ft$

$M_{precomp_g} := M_{SW_g} + 2\ V_{precomp_i} \cdot infill_spacing = 212.1\ kip \cdot ft$ $M_{precomp_g} < 221.2\ kip \cdot ft$

$\Delta_{SW_g} := \frac{5 \cdot girder_SW \cdot girder_span^4}{384 \cdot E_s \cdot I_{x_g}} = 0.0411\ in$

$\Delta_{con_g} := \Delta_{SW_g} + 2 \cdot \frac{23 \cdot V_{con_i} \cdot girder_span^3}{648 \cdot E_s \cdot I_{x_g}} = 0.9088\ in$

$camber_g := 0.8 \cdot \Delta_{con_g} = 0.727\ in$

$camber_g := 0.75\ in$ gets rounded to nearest 1/4"

$\Delta_{comp_LL_g} := 2 \cdot \frac{23 \cdot V_{comp_LL_i} \cdot girder_span^3}{648 \cdot E_s \cdot I_{LB_g}} = 0.28\ in$ $\Delta_{comp_LL_g} < \frac{girder_span}{360}$ or 1"

$\Delta_{comp_DL_g} := \Delta_{con_g} - camber_g = 0.1588\ in$ $\Delta_{comp_DL_g} < \frac{girder_span}{360}$ or 1"

$\Delta_{comp_TL_g} := \Delta_{comp_LL_g} + \Delta_{comp_DL_g} = 0.4389\ in$ $\Delta_{comp_TL_g} < \frac{girder_span}{240}$ or 1.5"

Vibration (Peak Acceleration) Calculation

$LL_{vib} := 6\ psf$

$DL_{vib} := 4\ psf$

$\beta := 0.02$

$$n := \frac{E_s}{1.35 \cdot E_c} = 9.31$$

$$conc_width_{eff_i} := min\left(0.4 \cdot infill_span, infill_spacing\right) = 120\ in$$

$$conc_width_{tran_i} := \frac{conc_width_{eff_i}}{n} = 12.89\ in$$

$$conc_depth_{eff} := topping_depth = 2\ in$$

$$conc_area_{tran_i} := conc_depth_{eff} \cdot conc_width_{tran_i} = 25.779\ in^2$$

$$y_{bar_i} := \frac{conc_area_{tran_i} \cdot \left(\frac{d_i}{2} + deck_depth + \frac{conc_depth_{eff}}{2}\right)}{conc_area_{tran_i} + A_i} = 7.483\ in$$

$$I_i := \frac{conc_width_{tran_i} \cdot conc_depth_{eff}^{\ 3}}{12} + conc_area_{tran_i} \cdot \left(\frac{d_i}{2} + deck_depth + \frac{conc_depth_{eff}}{2} - y_{bar_i}\right)^2 + I_{x_i} + A_i \cdot y_{bar_i}^{\ 2} = 517.891\ in^4$$

$$w_i := infill_spacing \cdot \left(LL_{vib} + deck_slab_SW + DL_{vib}\right) + infill_SW = 410\ plf$$

$$\Delta_i := \frac{5 \cdot w_i \cdot infill_span^4}{384 \cdot E_s \cdot I_i} = 0.498\ in$$

$$g := 386\ \frac{in}{s^2}$$

$$f_i := 0.18 \cdot \sqrt[2]{\frac{g}{\Delta_i}} = 5.014\ Hz$$

$$d_e := topping_depth + \frac{deck_depth}{2} = 3\ in$$

$$D_s := \frac{\left(\frac{12\ in}{1\ ft}\right) d_e^{\ 3}}{12\ n} = 2.9\ \frac{in^4}{ft}$$

$$D_i := \frac{I_i}{infill_spacing} = 51.789\ \frac{in^4}{ft}$$

$C_j := 2.0$

$B_i := C_j \cdot \left(\frac{D_s}{D_i}\right)^{0.25} \cdot infill_span = 29.187\ ft$

floor width and length are assumed to be large enough to not cause any problems

$W_i := 1.5 \cdot \left(\frac{w_i}{infill_spacing}\right) \cdot B_i \cdot infill_span = 53850.9\ lbf$

$conc_width_{eff_g} := 2 \cdot min\left(0.2 \cdot girder_span, 0.5 \cdot infill_span\right) = 144\ in$

$conc_width_{tran_g} := \frac{conc_width_{eff_g}}{n} = 15.467\ in$

$deck_width_{eff_g} := \frac{conc_width_{eff_g}}{2\ n} = 7.734\ in$

$conc_area_{tran_g} := conc_depth_{eff} \cdot conc_width_{tran_g} = 30.935\ in^2$

$deck_area_{tran_g} := deck_depth \cdot deck_width_{eff_g} = 15.467\ in^2$

$$y_{bar_g} := \frac{conc_area_{tran_g} \cdot \left(\frac{d_g}{2} + deck_depth + \frac{conc_depth_{eff}}{2}\right) + deck_area_{tran_g} \cdot \left(\frac{d_g}{2} + \frac{deck_depth}{2}\right)}{conc_area_{tran_g} + deck_area_{tran_g} + A_g} = 9\ in$$

$$I_g := \frac{conc_width_{tran_g} \cdot conc_depth_{eff}^{\ 3}}{12} + conc_area_{tran_g} \cdot \left(\frac{d_g}{2} + deck_depth + \frac{conc_depth_{eff}}{2} - y_{bar_g}\right)^2 + \frac{deck_width_{eff_g} \cdot deck_depth^3}{12} + deck_area_{tran_g} \cdot \left(\frac{d_g}{2} + \frac{deck_depth}{2} - y_{bar_g}\right)^2 + I_{x_g} + A_g \cdot y_{bar_g}^{\ 2} = 1866.4\ in^4$$

$w_g := infill_span \cdot \left(\frac{w_i}{infill_spacing}\right) + girder_SW = 1270\ plf$

$\Delta_g := \frac{5 \cdot w_g \cdot girder_span^4}{384 \cdot E_s \cdot I_g} = 0.428\ in$

$f_g := 0.18 \cdot \sqrt[2]{\frac{g}{\Delta_g}} = 5.408\ Hz$

$D_g := \frac{I_g}{infill_span} = 62.215\ \frac{in^4}{ft}$

$C_g := 1.8$

$B_g := C_g \cdot \left(\frac{D_i}{D_g}\right)^{0.25} \cdot girder_span = 51.58\ ft$

$W_g := \left(\frac{w_g}{infill_span}\right) \cdot B_g \cdot infill_span = 65506.5\ lbf$

$f_n := 0.18 \cdot \sqrt[2]{\frac{g}{\Delta_i + \Delta_g}} = 3.677\ Hz$

$\Delta_g' := \Delta_g = 0.428\ in$ Girder Span is > Bi

$W := \frac{\Delta_i}{\Delta_i + \Delta_g'} \cdot W_i + \frac{\Delta_g}{\Delta_i + \Delta_g'} \cdot W_g = 59238.3\ lbf$

$P_o := 65\ lbf$

$a_{p_over_g} := \left(\frac{P_o \cdot e^{-0.35 \cdot \frac{f_n}{Hz}}}{\beta \cdot W}\right) \cdot 100 = 1.51498\ \%g$

Embodied Carbon Calculation

$bay_area := girder_span \cdot infill_span = 900\ ft^2$

$W_{infill} := infill_SW \cdot infill_span \cdot (number_infills + 2) = 2280\ lbf$

$W_{girder} := girder_SW \cdot girder_span \cdot 2 = 2400\ lbf$

$W_{deck} := deck_SW \cdot bay_area = 1440\ lbf$

$W_{per_stud} := 0.437\ lbf$

$W_{studs_infill} := W_{per_stud} \cdot 17 \cdot (number_infills + 2) = 29.7\ lbf$

$W_{studs_girder} := W_{per_stud} \cdot 41 \cdot 2 = 35.8\ lbf$

$W_{concrete} := slab_SW \cdot bay_area = 24750\ lbf$

$EC_{steel_members} := 1.22 \cdot \left(\frac{W_{infill} + W_{girder}}{lbf} \right) = 5709.6$ lbCO2e/lb

$EC_{deck} := 2.32 \cdot \left(\frac{W_{deck}}{lbf} \right) = 3340.8$ lbCO2e/lb

$EC_{studs} := 1.27 \cdot \left(\frac{W_{studs_infill} + W_{studs_girder}}{lbf} \right) = 83.2$ lbCO2e/lb

$EC_{concrete} := 0.307 \cdot \left(\frac{W_{concrete}}{lbf} \right) = 7598.3$

$Scaled_EC := \frac{EC_{steel_members} + EC_{deck} + EC_{studs} + EC_{concrete}}{\frac{bay_area}{ft^2}} = 18.591$ lbCO2e/(lb*ft^2)

Spot Checks: Checking Slender Moment Capacity (Lr < Lb)

* For any of the 'Spot Checks', values highlighted in green match those output in code

To achieve slender conditions, the following design was used:
Girder Span: 32'
Infill Span: 20'
Equally Spaced Infills: 1
Concrete Type: LWC
Deck Depth: 3"
Percent Composite Connection: 50%
Occupancy: Residential

* Only the pre composite capacity of the W18x40 Girder selected in the code is checked to ensure Slender Moment Capacity is calculated properly

$L_b := \frac{32\ ft}{2} = 16\ ft$ (2 equal spaces)

$L_r := 15.6\ ft$

$L_b > L_r$ so the girder is slender

$C_b := 1.67$ (2 equal spaces Table 3-1)
$r_{ts} := 1.80\ in$
$J := 0.569\ in^4$
$S_x := 48.6\ in^3$
$h_o := 13.5\ in$

$$F_{cr} := \frac{C_b \cdot \pi^2 \cdot E_s}{\left(\frac{L_b}{r_{ts}}\right)^2} \cdot \sqrt{1 + 0.078 \cdot \frac{J}{S_x \cdot h_o} \left(\frac{L_b}{r_{ts}}\right)^2} = 55.886\ ksi$$

$\phi M_n := 0.9 \cdot S_x \cdot F_{cr} = 203.703\ kip \cdot ft$ (Matches value calculated in code)

Spot Checks: Making sure compact members aren't checked for LTB (Lp > Lb)

To achieve compact conditions, the following design was used:
Girder Span: 10'
Infill Span: 40'
Equally Spaced Infills: 4
Concrete Type: LWC
Deck Depth: 3"
Percent Composite Connection: 50%
Occupancy: Residential

* It is understood this is an unrealistic design, it is used to ensure the boundaries of my design space are being calculated correctly
* Only the flexural capacity of the W12x14 Girder selected in the code is checked to ensure Compact Moment Capacity is calculated properly

$L_b := \frac{10\ ft}{5} = 2\ ft$ (5 equal spaces)

$L_p := 2.87\ ft$

$L_b < L_p$ so the girder is compact, and PhiMpx from Table 3-2 is used

$\phi M_n := 47\ kip \cdot ft$ (Matches value used in code)

Spot Check: Partially Composite Case where PNA in Steel Beam Web

* In the full worked out example above, the PNA of the infill and girder is in the Steal Beam Flange, therefore the composite case where the PNA is in the Steel Beam Web must be checked.

To achieve PNA in the web for both members, the following design was used:
Girder Span: 30'
Infill Span: 17'
Equally Spaced Infills: 1
Concrete Type: NWC
Deck Depth: 3"
Percent Composite Connection: 25%
Occupancy: Residential

$girder_span := 30\ ft$
$infill_span := 17\ ft$
$number_infills := 1$
$PCC := 25\%$
$deck_depth := 3\ in$
$topping_depth := 2\ in$

Infill Composite Values
Selected Infill: W12x14 [7]
$E_s := 29000\ ksi$
$I_{x_i} := 89\ in^4$
$f'_c := 4\ ksi$
$A_i := 4.16\ in^2$
$F_y := 50\ ksi$
$F_u := 65\ ksi$
$b_{f_i} := 3.97\ in$
$d_i := 11.9\ in$
$t_{w_i} := 0.2\ in$
$R_g := 1$
$R_p := 0.6$
$w_c := 145\ pcf$

$$A_{stud} := \frac{\pi \cdot (0.75\ in)^2}{4} = 0.442\ in^2$$

$$E_c := \left(\frac{w_c}{pcf}\right)^{1.5} \cdot \sqrt[2]{\frac{f'_c}{ksi}} = 3.492 \cdot 10^3 \qquad E_c := 2307.38\ ksi$$

$$infill_spacing := \frac{girder_span}{number_infills + 1} = 15\ ft$$

$b_{eff_i} := min\left(\frac{infill_span \cdot 2}{8}, \frac{infill_spacing \cdot 2}{2}\right) = 51\ in$

$V_{c_i}' := 0.85 \cdot f'_c \cdot b_{eff_i} \cdot topping_depth = 346.8\ kip$

$V_{s_i}' := A_i \cdot F_y = 208\ kip$

$V_i' := min\left(V_{c_i}', V_{s_i}'\right) = 208\ kip$

$Q_{n_i} := min\left(0.5 \cdot A_{stud} \cdot \sqrt[2]{\left(f'_c \cdot E_c\right)}, R_g \cdot R_p \cdot A_{stud} \cdot F_u\right) = 17.23\ kip$

$\Sigma Q_{n_i} := PCC \cdot V_i' = 52\ kip$

7 studs are specified, therefore 3.5 studs * Qn = 60.3kip > 52kip

$a_i := \frac{\Sigma Q_{n_i}}{0.85 \cdot f'_c \cdot b_{eff_i}} = 0.3\ in$

$C_{c_i} := 0.85 \cdot f'_c \cdot b_{eff_i} \cdot a_i = 52\ kip$

$t_{f_i} := 0.025\ in$

$Y1_i := \frac{d_i}{2} - \frac{C_{c_i}}{2\ t_{w_i} \cdot F_y} = 3.35\ in$

Y1 > tf, PNA is in the Steel Beam Web

$Y2_i := deck_depth + \frac{topping_depth}{2} = 4\ in$

$T_i := A_i \cdot F_y = 208\ kip$

$T_{s1_i} := 2 \cdot t_{w_i} \cdot F_y \cdot \left(\frac{d_i}{2} - Y1_i\right) = 52\ kip$

$Z_i := 17.4\ in^3$

$M_{p_i} := F_y \cdot Z_i$

$\phi M_{n_i} := 0.9 \cdot \left(C_{c_i} \cdot \left(Y2_i + \frac{d_i}{2}\right) - T_{s1_i} \cdot \left(\frac{\frac{d_i}{2} - Y1_i}{2}\right) + M_{p_i}\right) = 98.985\ kip \cdot ft$

$d_{1_i} := topping_depth + deck_depth - \frac{a_i}{2} = 4.85\ in$

$d_{3_i} := \frac{d_i}{2} = 5.95\ in$

$Y_{ENA_i} := \frac{\left(A_i \cdot d_{3_i} + \frac{\Sigma Q_{n_i}}{F_y} \cdot \left(2 \cdot d_{3_i} + d_{1_i}\right)\right)}{\left(A_i + \frac{\Sigma Q_{n_i}}{F_y}\right)} = 8.11\ in$

$I_{LB_i} := I_{x_i} + A_i \cdot \left(Y_{ENA_i} - d_{3_i}\right)^2 + \frac{\Sigma Q_{n_i}}{F_y} \cdot \left(2 \cdot d_{3_i} + d_{1_i} - Y_{ENA_i}\right)^2 = 186.046\ in^4$

Girder Composite Values

Selected Infill: W18x35 [15]

$I_{x_g} := 510\ in^4$

$A_g := 10.3\ in^2$

$b_{f_g} := 6.00\ in$

$d_g := 17.7\ in$

$t_{w_g} := 0.3\ in$

$$b_{eff_g} := min\left(\frac{girder_span \cdot 2}{8}, \frac{infill_span}{2}\right) = 90\ in$$

$$V_{c_g}' := 0.85 \cdot f'_c \cdot b_{eff_g} \cdot topping_depth = 612\ kip$$

$$V_{s_g}' := A_g \cdot F_y = 515\ kip$$

$$V_g' := min\left(V_{c_g}', V_{s_g}'\right) = 515\ kip$$

$$Q_{n_g} := min\left(0.5 \cdot A_{stud} \cdot \sqrt[2]{f'_c \cdot E_c}, R_g \cdot R_p \cdot A_{stud} \cdot F_u\right) = 17.23\ kip$$

$$\Sigma Q_{n_g} := PCC \cdot V_g' = 128.75\ kip$$

15 studs are specified, therefore 7.5 studs * Qn = 129.23kip > 128.75kip

$$a_g := \frac{\Sigma Q_{n_g}}{0.85 \cdot f'_c \cdot b_{eff_g}} = 0.421\ in$$

$$C_{c_g} := 0.85 \cdot f'_c \cdot b_{eff_g} \cdot a_g = 128.75\ kip$$

$t_{f_g} := 0.025\ in$

$$Y1_g := \frac{d_g}{2} - \frac{C_{c_g}}{2\ t_{w_g} \cdot F_y} = 4.558\ in$$

Y1 > tf, PNA is in the Steel Beam Web

$$Y2_g := deck_depth + \frac{topping_depth}{2} = 4\ in$$

$$T_g := A_g \cdot F_y = 515\ kip$$

$$T_{s1_g} := 2 \cdot t_{w_g} \cdot F_y \cdot \left(\frac{d_g}{2} - Y1_g\right) = 128.75\ kip$$

$Z_g := 66.5\ in^3$

$M_{p_g} := F_y \cdot Z_g$

$$\phi M_{n_g} := 0.9 \cdot \left(C_{c_g} \cdot \left(Y2_g + \frac{d_g}{2}\right) - T_{s1_g} \cdot \left(\frac{\frac{d_g}{2} - Y1_g}{2}\right) + M_{p_g}\right) = 352.737\ kip \cdot ft$$

$$d_{1_g} := topping_depth + deck_depth - \frac{a_g}{2} = 4.79\ in$$

$$d_{3_g} := \frac{d_g}{2} = 8.85\ in$$

$$Y_{ENA_g} := \frac{\left(A_g \cdot d_{3_g} + \frac{\Sigma Q_{n_g}}{F_y} \cdot \left(2 \cdot d_{3_g} + d_{1_g}\right)\right)}{\left(A_g + \frac{\Sigma Q_{n_g}}{F_y}\right)} = 11.578\ in$$

$$I_{LB_g} := I_{x_g} + A_g \cdot \left(Y_{ENA_g} - d_{3_g}\right)^2 + \frac{\Sigma Q_{n_g}}{F_y} \cdot \left(2 \cdot d_{3_g} + d_{1_g} - Y_{ENA_g}\right)^2 = 893.241\ in^4$$

Spot Checks: Shear, Moment, and Deflection Values for One Infill

* Used Same Design as previous spot check

$LL := 40\ psf$

$SDL := 20\ psf$

3" Deck 16Ga w/ 2" NWC

$Deck_Slab_SW := 45.8\ psf$

$DL := SDL + Deck_Slab_SW = 65.8\ psf$

Infill Values:

$infill_SW := 14\ plf$ (Infill Self Weight)

$w_{DL_i} := DL \cdot infill_spacing + infill_SW = 1.001\ klf$ (Infill DL Uniform Load)

$w_{LL_i} := LL \cdot infill_spacing$ (Infill LL Uniform Load)

$w_{FL_i} := 1.2 \cdot w_{DL_i} + 1.6 \cdot w_{LL_i} = 2.161\ klf$ (Infill Factored Uniform Load)

(Precomposite Factored Uniform Load)

$w_{precomp_i} := 1.2 \cdot (Deck_Slab_SW \cdot infill_spacing + infill_SW) + 1.6\ (20\ psf \cdot infill_spacing) = 1.321\ klf$

$w_{con_i} := Deck_Slab_SW \cdot infill_spacing + infill_SW = 0.701\ klf$

$w_{comp_LL_i} := LL \cdot infill_spacing \cdot 0.5 = 0.3\ klf$

$$V_{DL_i} := \frac{w_{DL_i} \cdot infill_span}{2} = 8.509\ kip$$

$$V_{LL_i} := \frac{w_{LL_i} \cdot infill_span}{2} = 5.1\ kip$$

$$V_{FL_i} := \frac{w_{FL_i} \cdot infill_span}{2} = 18.37\ kip \qquad V_{FL_i} < 71\ kip$$

$$V_{con_i} := \frac{w_{con_i} \cdot infill_span}{2} = 5.959\ kip$$

$$V_{precomp_i} := \frac{w_{precomp_i} \cdot infill_span}{2} = 11.23\ kip$$

$$V_{comp_LL_i} := \frac{w_{comp_LL_i} \cdot infill_span}{2} = 2.55\ kip$$

$$M_{FL_i} := \frac{w_{FL_i} \cdot infill_span^2}{8} = 78.073\ kip \cdot ft \qquad M_{FL_i} < 99\ kip \cdot ft$$

$$M_{precomp_i} := \frac{w_{precomp_i} \cdot infill_span^2}{8} = 47.728\ kip \cdot ft \qquad M_{precomp_i} < 65\ kip \cdot ft$$

$$\Delta_{con_i} := \frac{5 \cdot w_{con_i} \cdot infill_span^4}{384 \cdot E_s \cdot I_{x_i}} = 0.5104\ in$$

$camber := 0.8 \cdot \Delta_{con_i} = 0.408\ in$

$camber_i := 0\ in$ must be at least 0.75"

$\Delta_{comp_LL_i} := \frac{5 \cdot w_{comp_LL_i} \cdot infill_span^4}{384 \cdot E_s \cdot I_{LB_i}} = 0.1045\ in$ $\Delta_{comp_LL_g} < \frac{girder_span}{360}$ or 1"

$\Delta_{comp_DL_i} := \Delta_{con_i} - camber_i = 0.5104\ in$ $\Delta_{comp_DL_g} < \frac{girder_span}{360}$ or 1"

$\Delta_{comp_TL_i} := \Delta_{comp_LL_i} + \Delta_{comp_DL_i} = 0.6149\ in$ $\Delta_{comp_TL_g} < \frac{girder_span}{240}$ or 1.5"

Girder Values:

$girder_SW := 35\ plf$ (Girder Self Weight)

$I_{x_g} := 510\ in^4$

$V_{DL_g} := \frac{(girder_SW \cdot girder_span + 2 \cdot V_{DL_i} \cdot number_infills)}{2} = 9.034\ kip$

$V_{LL_g} := \frac{2 \cdot V_{LL_i} \cdot number_infills}{2} = 5.1\ kip$

$V_{FL_g} := 1.2 \cdot V_{DL_g} + 1.6 \cdot V_{LL_g} = 19\ kip$ $V_{FL_g} < 159\ k$

$V_{precomp_g} := \frac{(girder_SW \cdot girder_span + 2 \cdot V_{precomp_i} \cdot number_infills)}{2} = 11.755\ kip$

$V_{con_g} := \frac{(girder_SW \cdot girder_span + 2 \cdot V_{con_i} \cdot number_infills)}{2} = 6.484\ kip$

$M_{SW_g} := \frac{girder_SW \cdot girder_span^2}{8} = 3.938\ kip \cdot ft$

$M_{DL_g} := M_{SW_g} + 2\ \frac{V_{DL_i} \cdot girder_span}{4} = 131.565\ kip \cdot ft$

$M_{LL_g} := 2\ \frac{V_{LL_i} \cdot girder_span}{4} = 76.5\ kip \cdot ft$

$M_{FL_g} := 1.2 \cdot M_{DL_g} + 1.6 \cdot M_{LL_g} = 280.278\ kip \cdot ft$ $M_{FL_g} < 352.7\ kip \cdot ft$

$M_{precomp_g} := M_{SW_g} + 2\ \frac{V_{precomp_i} \cdot girder_span}{4} = 172.391\ kip \cdot ft$ $M_{precomp_g} < 181.7\ kip \cdot ft$

$\Delta_{SW_g} := \frac{5 \cdot girder_SW \cdot girder_span^4}{384 \cdot E_s \cdot I_{x_g}} = 0.0431\ in$

$\Delta_{con_g} := \Delta_{SW_g} + 2 \cdot \frac{V_{con_i} \cdot girder_span^3}{48 \cdot E_s \cdot I_{x_g}} = 0.8263\ in$

$camber_g := 0.8 \cdot \Delta_{con_g} = 0.661\ in$

$camber_g := 0.75\ in$ gets rounded to nearest 1/4"

$\Delta_{comp_LL_g} := 2 \cdot \frac{V_{comp_LL_i} \cdot girder_span^3}{48 \cdot E_s \cdot I_{LB_g}} = 0.1914\ in$ $\Delta_{comp_LL_g} < \frac{girder_span}{360}$ or 1"

$\Delta_{comp_DL_g} := \Delta_{con_g} - camber_g = 0.0763\ in$ $\Delta_{comp_DL_g} < \frac{girder_span}{360}$ or 1"

$\Delta_{comp_TL_g} := \Delta_{comp_LL_g} + \Delta_{comp_DL_g} = 0.2677\ in$ $\Delta_{comp_TL_g} < \frac{girder_span}{240}$ or 1.5"

Spot Checks: Shear, Moment, and Deflection Values for Three Infills

The following design was used:
Girder Span: 30'
Infill Span: 30'
Equally Spaced Infills: 3
Concrete Type: NWC
Deck Depth: 3"
Percent Composite Connection: 50%
Occupancy: Residential

$infill_span := 30\ ft$
$girder_span := 30\ ft$

$infill_spacing := \frac{girder_span}{4} = 7.5\ ft$

$LL := 40\ psf$
$SDL := 20\ psf$
3" Deck 22Ga w/ 2" NWC
$Deck_Slab_SW := 44\ psf$
$DL := SDL + Deck_Slab_SW = 64\ psf$

Infill Values:
Infill Size: W12x16 (14) $I_{x_i} := 103\ in^4$ $I_{LB_i} := 286.38\ in^4$
$infill_SW := 16\ plf$ (Infill Self Weight)
$w_{DL_i} := DL \cdot infill_spacing + infill_SW = 0.496\ klf$ (Infill DL Uniform Load)
$w_{LL_i} := LL \cdot infill_spacing$ (Infill LL Uniform Load)
$w_{FL_i} := 1.2 \cdot w_{DL_i} + 1.6 \cdot w_{LL_i} = 1.075\ klf$ (Infill Factored Uniform Load)
(Precomposite Factored Uniform Load)
$w_{precomp_i} := 1.2 \cdot (Deck_Slab_SW \cdot infill_spacing + infill_SW) + 1.6\ (20\ psf \cdot infill_spacing) = 0.655\ klf$
$w_{con_i} := Deck_Slab_SW \cdot infill_spacing + infill_SW = 0.346\ klf$
$w_{comp_LL_i} := LL \cdot infill_spacing \cdot 0.5 = 0.15\ klf$

$V_{DL_i} := \frac{w_{DL_i} \cdot infill_span}{2} = 7.44\ kip$

$V_{LL_i} := \frac{w_{LL_i} \cdot infill_span}{2} = 4.5\ kip$

$V_{FL_i} := \frac{w_{FL_i} \cdot infill_span}{2} = 16.128\ kip$ $V_{FL_i} < 79\ kip$

$V_{con_i} := \frac{w_{con_i} \cdot infill_span}{2} = 5.19\ kip$

$V_{precomp_i} := \frac{w_{precomp_i} \cdot infill_span}{2} = 9.828\ kip$

$V_{comp_LL_i} := \frac{w_{comp_LL_i} \cdot infill_span}{2} = 2.25\ kip$

$M_{FL_i} := \frac{w_{FL_i} \cdot infill_span^2}{8} = 120.96\ kip \cdot ft$ $M_{FL_i} < 140.1\ kip \cdot ft$

$M_{precomp_i} := \frac{w_{precomp_i} \cdot infill_span^2}{8} = 73.71\ kip \cdot ft$ $M_{precomp_i} < 75\ kip \cdot ft$

$\Delta_{con_i} := \frac{5 \cdot w_{con_i} \cdot infill_span^4}{384 \cdot E_s \cdot I_{x_i}} = 2.1111\ in$

$camber := 0.8 \cdot \Delta_{con_i} = 1.689\ in$

$camber_i := 1.75\ in$ rounded to nearest 1/4"

$\Delta_{comp_LL_i} := \frac{5 \cdot w_{comp_LL_i} \cdot infill_span^4}{384 \cdot E_s \cdot I_{LB_i}} = 0.3292\ in$ $\Delta_{comp_LL_g} < \frac{girder_span}{360}$ or 1"

$\Delta_{comp_DL_i} := \Delta_{con_i} - camber_i = 0.3611\ in$ $\Delta_{comp_DL_g} < \frac{girder_span}{360}$ or 1"

$\Delta_{comp_TL_i} := \Delta_{comp_LL_i} + \Delta_{comp_DL_i} = 0.6903\ in$ $\Delta_{comp_TL_g} < \frac{girder_span}{240}$ or 1.5"

Girder Values:

Girder Size: W21x44 (42)

$girder_SW := 44\ plf$ (Girder Self Weight)

$I_{x_g} := 843\ in^4$ $I_{LB_g} := 1760.61\ in^4$

$V_{DL_g} := \frac{girder_SW \cdot girder_span}{2} + 2 \cdot \frac{V_{DL_i} \cdot 3}{2} = 22.98\ kip$

$V_{LL_g} := 2 \cdot \frac{V_{LL_i} \cdot 3}{2} = 13.5\ kip$

$V_{FL_g} := 1.2 \cdot V_{DL_g} + 1.6 \cdot V_{LL_g} = 49.176\ kip$ $V_{FL_g} < 217\ k$

$V_{precomp_g} := \frac{girder_SW \cdot girder_span}{2} + 2 \cdot \frac{V_{precomp_i} \cdot 3}{2} = 30.144\ kip$

$V_{con_g} := \frac{girder_SW \cdot girder_span}{2} + 2 \cdot \frac{V_{con_i} \cdot 3}{2} = 16.23\ kip$

$M_{SW_g} := \frac{girder_SW \cdot girder_span^2}{8} = 4.95\ kip \cdot ft$

$M_{DL_g} := M_{SW_g} + 2 \cdot \frac{V_{DL_i} \cdot girder_span}{2} = 228.15\ kip \cdot ft$

$M_{LL_g} := 2 \cdot \frac{V_{LL_i} \cdot girder_span}{2} = 135\ kip \cdot ft$

$M_{FL_g} := 1.2 \cdot M_{DL_g} + 1.6 \cdot M_{LL_g} = 489.78\ kip \cdot ft$ $M_{FL_g} < 586.8\ kip \cdot ft$

$M_{precomp_g} := M_{SW_g} + 2 \cdot \frac{V_{precomp_i} \cdot girder_span}{2} = 299.79\ kip \cdot ft$ $M_{precomp_g} < 340.5\ kip \cdot ft$

$\Delta_{SW_g} := \frac{5 \cdot girder_SW \cdot girder_span^4}{384 \cdot E_s \cdot I_{x_g}} = 0.0328\ in$

$\Delta_{con_g} := \Delta_{SW_g} + 2\ \frac{19\ V_{con_i} \cdot girder_span^3}{384 \cdot E_s \cdot I_{x_g}} = 1.013\ in$

$camber_g := 0.8 \cdot \Delta_{con_g} = 0.81\ in$

$camber_g := 0.75\ in$ gets rounded to nearest 1/4"

$\Delta_{comp_LL_g} := 2\ \frac{19\ V_{comp_LL_i} \cdot girder_span^3}{384 \cdot E_s \cdot I_{LB_g}} = 0.2035\ in$ $\Delta_{comp_LL_g} < \frac{girder_span}{360}$ or 1"

$\Delta_{comp_DL_g} := \Delta_{con_g} - camber_g = 0.263\ in$ $\Delta_{comp_DL_g} < \frac{girder_span}{360}$ or 1"

$\Delta_{comp_TL_g} := \Delta_{comp_LL_g} + \Delta_{comp_DL_g} = 0.4664\ in$ $\Delta_{comp_TL_g} < \frac{girder_span}{240}$ or 1.5"

Spot Checks: Shear, Moment, and Deflection Values for Four Infills

The following design was used:
Girder Span: 30'
Infill Span: 30'
Equally Spaced Infills: 4
Concrete Type: NWC
Deck Depth: 3"
Percent Composite Connection: 50%
Occupancy: Residential

$infill_span := 30\ ft$
$girder_span := 30\ ft$

$infill_spacing := \frac{girder_span}{5} = 6\ ft$

$LL := 40\ psf$
$SDL := 20\ psf$
3" Deck 22Ga w/ 2" NWC
$Deck_Slab_SW := 44\ psf$
$DL := SDL + Deck_Slab_SW = 64\ psf$

Infill Values:
Infill Size: W12x14 (13) $I_{x_i} := 89\ in^4$ $I_{LB_i} := 248.877\ in^4$
$infill_SW := 14\ plf$ (Infill Self Weight)
$w_{DL_i} := DL \cdot infill_spacing + infill_SW = 0.398\ klf$ (Infill DL Uniform Load)
$w_{LL_i} := LL \cdot infill_spacing$ (Infill LL Uniform Load)
$w_{FL_i} := 1.2 \cdot w_{DL_i} + 1.6 \cdot w_{LL_i} = 0.862\ klf$ (Infill Factored Uniform Load)
(Precomposite Factored Uniform Load)
$w_{precomp_i} := 1.2 \cdot (Deck_Slab_SW \cdot infill_spacing + infill_SW) + 1.6\ (20\ psf \cdot infill_spacing) = 0.526\ klf$
$w_{con_i} := Deck_Slab_SW \cdot infill_spacing + infill_SW = 0.278\ klf$
$w_{comp_LL_i} := LL \cdot infill_spacing \cdot 0.5 = 0.12\ klf$

$V_{DL_i} := \frac{w_{DL_i} \cdot infill_span}{2} = 5.97\ kip$

$V_{LL_i} := \frac{w_{LL_i} \cdot infill_span}{2} = 3.6\ kip$

$V_{FL_i} := \frac{w_{FL_i} \cdot infill_span}{2} = 12.924\ kip$ $V_{FL_i} < 71\ kip$

$V_{con_i} := \frac{w_{con_i} \cdot infill_span}{2} = 4.17\ kip$

$V_{precomp_i} := \frac{w_{precomp_i} \cdot infill_span}{2} = 7.884\ kip$

$V_{comp_LL_i} := \frac{w_{comp_LL_i} \cdot infill_span}{2} = 1.8\ kip$

$M_{FL_i} := \frac{w_{FL_i} \cdot infill_span^2}{8} = 96.93\ kip \cdot ft$ $M_{FL_i} < 122.6\ kip \cdot ft$

$M_{precomp_i} := \frac{w_{precomp_i} \cdot infill_span^2}{8} = 59.13\ kip \cdot ft$ $M_{precomp_i} < 65\ kip \cdot ft$

$\Delta_{con_i} := \frac{5 \cdot w_{con_i} \cdot infill_span^4}{384 \cdot E_s \cdot I_{x_i}} = 1.963\ in$

$camber := 0.8 \cdot \Delta_{con_i} = 1.57\ in$

$camber_i := 1.5\ in$ rounded to nearest 1/4"

$\Delta_{comp_LL_i} := \frac{5 \cdot w_{comp_LL_i} \cdot infill_span^4}{384 \cdot E_s \cdot I_{LB_i}} = 0.303\ in$ $\Delta_{comp_LL_g} < \frac{girder_span}{360}$ or 1"

$\Delta_{comp_DL_i} := \Delta_{con_i} - camber_i = 0.463\ in$ $\Delta_{comp_DL_g} < \frac{girder_span}{360}$ or 1"

$\Delta_{comp_TL_i} := \Delta_{comp_LL_i} + \Delta_{comp_DL_i} = 0.766\ in$ $\Delta_{comp_TL_g} < \frac{girder_span}{240}$ or 1.5"

Girder Values:

Girder Size: W21x44 (42)

$girder_SW := 44\ plf$

$I_{x_g} := 843\ in^4$ $I_{LB_g} := 1760.61\ in^4$

$w_{DL_g} := DL \cdot infill_span = 1.92\ klf$

$w_{LL_g} := LL \cdot infill_span = 1.2\ klf$ (Girder Self Weight)

$w_{SW_i} := \frac{4 \cdot infill_SW \cdot infill_span}{girder_span} = 0.056\ klf$ (Self weight of all infills resting on girder)

$w_{precomp_g} := 1.2 \cdot (Deck_Slab_SW \cdot infill_span + w_{SW_i} + girder_SW) + 1.6 \cdot (20\ psf \cdot infill_span) = 2.664\ klf$

$w_{con_g} := Deck_Slab_SW \cdot infill_span + w_{SW_i} = 1.376\ klf$

$w_{comp_LL_g} := LL \cdot infill_span \cdot 0.5 = 0.6\ klf$

$V_{DL_g} := \frac{girder_SW \cdot girder_span}{2} + 2 \cdot \frac{V_{DL_i} \cdot 4}{2} = 24.54\ kip$

$V_{LL_g} := 2 \cdot \frac{V_{LL_i} \cdot 4}{2} = 14.4\ kip$

$V_{FL_g} := 1.2 \cdot V_{DL_g} + 1.6 \cdot V_{LL_g} = 52.488\ kip$ $V_{FL_i} < 217\ kip$

$V_{precomp_g} := \frac{girder_SW \cdot girder_span}{2} + 2 \cdot \frac{V_{precomp_i} \cdot 4}{2} = 32.196\ kip$

$V_{con_g} := \frac{girder_SW \cdot girder_span}{2} + 2 \cdot \frac{V_{con_i} \cdot 4}{2} = 17.34\ kip$

$M_{DL_g} := \frac{(girder_SW + w_{SW_i} + w_{DL_g}) \cdot girder_span^2}{8} = 227.25\ kip \cdot ft$

$M_{LL_g} := \frac{w_{LL_g} \cdot girder_span^2}{8} = 135\ kip \cdot ft$

$M_{FL_g} := 1.2 \cdot M_{DL_g} + 1.6 \cdot M_{LL_g} = 488.7\ ft \cdot kip$ $\quad M_{FL_g} < 586.8\ kip \cdot ft$

$M_{precomp_g} := \frac{w_{precomp_g} \cdot girder_span^2}{8} = 299.7\ kip \cdot ft$ $\quad M_{precomp_g} < 332\ kip \cdot ft$

$\Delta_{SW_g} := \frac{5 \cdot girder_SW \cdot girder_span^4}{384 \cdot E_s \cdot I_{x_g}} = 0.0328\ in$

$\Delta_{con_g} := \Delta_{SW_g} + \frac{5 \cdot w_{con_g} \cdot girder_span^4}{384 \cdot E_s \cdot I_{x_g}} = 1.0586\ in$

$camber_g := 0.8 \cdot \Delta_{con_g} = 0.847\ in$

$camber_g := 0.75\ in$ gets rounded to nearest 1/4"

$\Delta_{comp_LL_g} := \frac{5 \cdot w_{comp_LL_g} \cdot girder_span^4}{384 \cdot E_s \cdot I_{LB_g}} = 0.2142\ in$ $\quad \Delta_{comp_LL_g} < \frac{girder_span}{360}$ or 1"

$\Delta_{comp_DL_g} := \Delta_{con_g} - camber_g = 0.3086\ in$ $\quad \Delta_{comp_DL_g} < \frac{girder_span}{360}$ or 1"

$\Delta_{comp_TL_g} := \Delta_{comp_LL_g} + \Delta_{comp_DL_g} = 0.5228\ in$ $\quad \Delta_{comp_TL_g} < \frac{girder_span}{240}$ or 1.5"